

Quantum Computation and Information

Lecture 6: Variational Quantum Algorithms

- variational principle
- variational quantum algorithms
 - overview
 - max-cut problem as an example
 - key issues

Hamiltonian

- Hamiltonian is self-adjoint and governs evolution of (closed) quantum system (Schrödinger equation)

$$\frac{d}{dt} |\psi(t)\rangle = -iH |\psi(t)\rangle$$

- eigenvalues E_i are called energies in physics

$$H |\psi_i\rangle = E_i |\psi_i\rangle, \quad i = 0, 1, 2, \dots, N-1$$

$$H = \sum_{i=0}^{N-1} E_i |\psi_i\rangle \langle \psi_i|$$

- ground state (i.e., state with lowest energy) is of particular interest, e.g.,
 - ground state exhibits exotic properties such as superconductivity in strongly correlated systems
 - ground state energy is important in quantum chemistry
- ground state and energy are hard to compute even if H is known

variational principle

- energy of state $|\psi\rangle$

$$E = \langle \psi | H | \psi \rangle = \sum_{i=0}^{N-1} E_i |\langle \psi_i | \psi \rangle|^2 \geq E_0 \sum_{i=0}^{N-1} |\langle \psi_i | \psi \rangle|^2 = E_0$$

– search over $|\psi\rangle$ can find E_0 , but impractical

- ansatz: try an educated guess $|\psi(\theta)\rangle$, parameterized by θ

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle \geq E_0$$

- minimize $E(\theta)$ over θ to obtain a good approximate of E_0 and $|\psi_0\rangle$

variational quantum algorithms

- provide a general framework for solving a variety of problems
 - given a problem and possibly some (training) data
 - construct a cost function C encoding the solution to the problem
 - propose an ansatz: unitary operation $U(\theta)$ by certain parameterized quantum circuit
 - * e.g., $|\psi(\theta)\rangle = U(\theta) |0\rangle$ for finding ground state energy/state
 - hybrid quantum–classical loop to solve for the optimum

$$\theta^* = \min_{\theta} C(\theta)$$

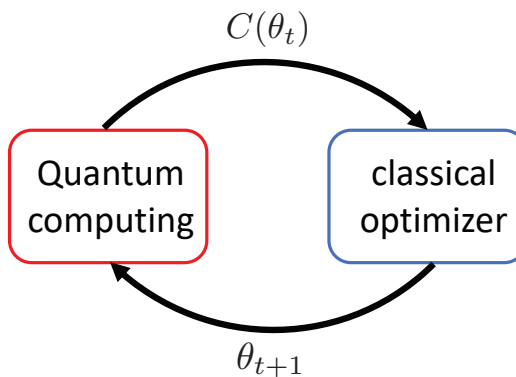
- * $C(\theta)$ or its derivative evaluated using **quantum** computer plus some possible classical post-processing
- * parameter θ updated using **classical** optimization algorithms (usually iterative algorithms)

variational quantum algorithms

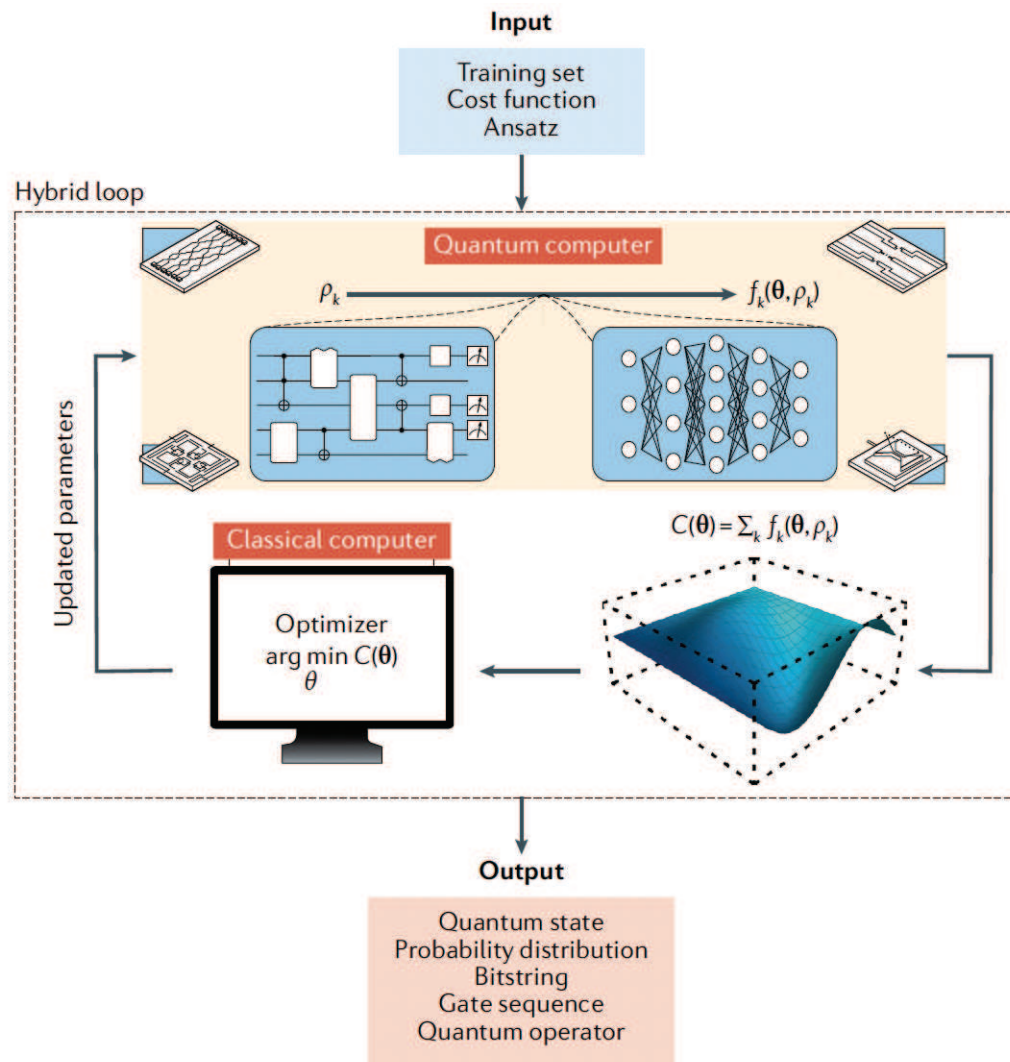
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variational quantum algorithms



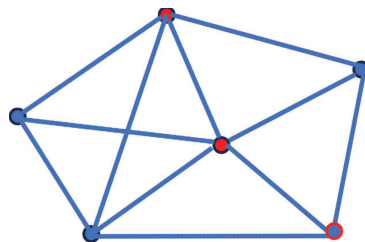
(Cerezo *et al* '21)

max-cut problem

- given a weighted graph $G = \{V, E, w\}$, find a cut $(U, V - U)$ maximizing

$$\sum_{i \in U, j \in V - U, (i, j) \in E} w_{ij}$$

- an example graph with edge weights $w_{ij} = 1$



- a quadratic binary optimization problem
 - introduce variable $x_i \in \{0, 1\}$ for each vertex $i \in V$ to indicate the partition into two complementary sets
 - let $w_{ij} = w_{ji}$, $w_{ii} = 0$, and $w_{ij} = 0$ for $(i, j) \notin E$
 - max-cut problem can be written as

$$\max_{x_i \in \{0, 1\}} C(x) = \sum_{i, j \in V} w_{ij} x_i (1 - x_j)$$

- an NP-hard problem
 - best classical polynomial time approximate algorithm achieves an approximation ratio of 0.87856

quantum approximate optimization algorithm

- encode the problem quantumly
 - encode the decision variables $x \in \{0, 1\}^n$ as quantum state $|x\rangle$
 - encode the cost function into a cost Hamiltonian H_c such that

$$H_c |x\rangle = C(x) |x\rangle$$

with $H_c = \frac{1}{4} \sum_{i,j} w_{ij} (I - Z_i Z_j)$

* Pauli-Z matrix $Z_i = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

* note that $\frac{1}{2}(I - Z_i) |x_i\rangle = x_i |x_i\rangle$

quantum approximate optimization algorithm

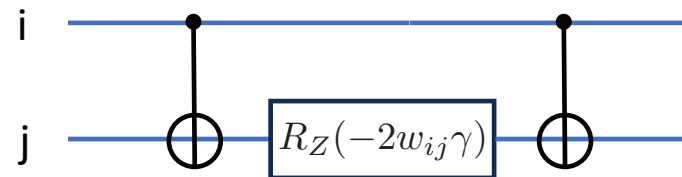
- ansatz

- introduce mixer Hamiltonian $H_m = \sum_i X_i$ with Pauli-X matrix

$$X_i = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

- define unitaries $U_c(\gamma) = e^{-iH_c\gamma}$ and $U_m(\beta) = e^{-iH_m\beta}$

- * $U_c(\gamma)$ implemented through applying to each pair (i, j) of interacting qubits the circuit



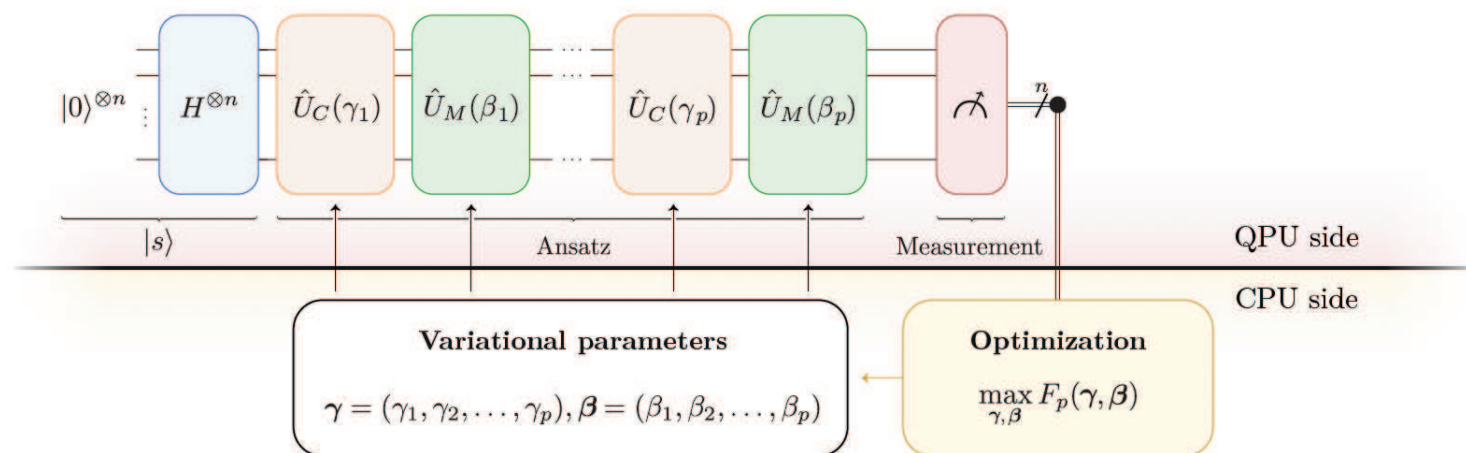
with rotation gate

$$R_Z(-2w_{ij}\gamma) = \begin{pmatrix} e^{iw_{ij}\gamma} & 0 \\ 0 & e^{-iw_{ij}\gamma} \end{pmatrix}$$

- * $U_m(\beta)$ implemented as applying rotation gate

$$R_X(2\beta) = \begin{pmatrix} \cos \beta & -i \sin \beta \\ i \sin \beta & \cos \beta \end{pmatrix} \text{ to each qubit}$$

- initialize to state $|s\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle$
- apply $U_c(\gamma_k)$ (cost layer) and $U_m(\beta_k)$ (mix layer) repeatedly



(Blekos *et al* '23)

- * a layered structure, each layer consisting of a cost (sub)layer and a mixer (sub)layer
- * partly motivated by quantum adiabatic computing
- * the final state after p layers

$$|\psi_p(\gamma, \beta)\rangle = e^{-i\beta_p H_m} e^{-i\gamma_p H_c} \dots e^{-i\beta_1 H_m} e^{-i\gamma_1 H_c} |s\rangle$$

- cost function (the expected value of H_c can be calculated based on repeated measurement of final state on computational basis)

$$C_p(\gamma, \beta) = \langle \psi_p(\gamma, \beta) | H_c | \psi_p(\gamma, \beta) \rangle$$

- classical optimization algorithm is used to iteratively update γ, β such that the cost function is maximized

$$(\gamma^*, \beta^*) = \arg \max_{\gamma, \beta} C_p(\gamma, \beta)$$

- $C_p(\gamma^*, \beta)^*$ is the ‘best’ approximation of $\max_x C(c)$
- $O(n^2p)$ complexity (versus $O(nm)$ for classical approximation algorithms)
- solution quality: not better than classical algorithms in general

variational quantum algorithms

data embedding and cost function

- embedding classical decision variables and data quantumly
 - nontrivial in general (and for quantum algorithms in general)
 - often rely on specific structure in the data, which may be also exploited by classical algorithms too
 - straightforward for certain problems; e.g., binary optimization problems
- cost function design

$$C(\theta) = f(\{\rho_k\}, \{O_k\}, U(\theta))$$

with input states ρ_k , observables O_k and parameterized circuit $U(\theta)$

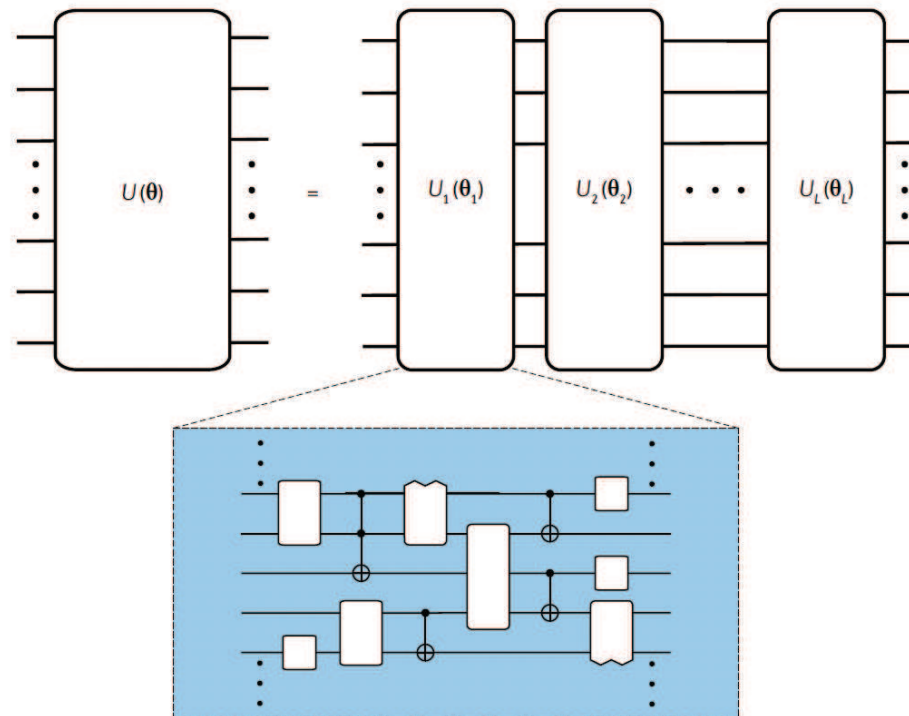
- usually problem specific, often come from the problem itself
- usually structured; e.g., additive with each terms involving few qubits
- be able to be efficiently estimated

variational quantum algorithms

ansatz

$$U(\theta) = U_L(\theta_L) \cdots U_2(\theta_2)U_1(\theta_1), \quad U_l(\theta_l) = \prod_m e^{i\theta_m^l H_m^l W_m^l}$$

- W_m^l 'unparameterized' unitary and H_m^l self-adjoint operator



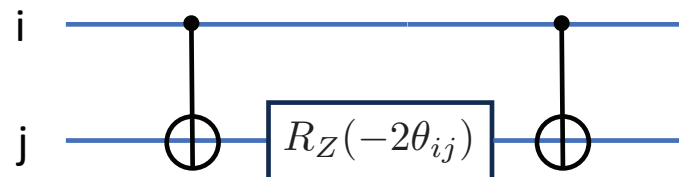
(Cerezo *et al* '21)

variational quantum algorithms

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- W_m^l 'unparameterized' unitary and H_m^l self-adjoint operator
- H_m^l is often well-structured; e.g., additive of single Pauli terms and pairwise Pauli terms
 - $e^{-i\theta_i X_i}$ as rotation gate $R_X(2\theta) = \begin{pmatrix} \cos \theta_i & -i \sin \theta_i \\ i \sin \theta_i & \cos \theta_i \end{pmatrix}$
 - $e^{-i\theta_{ij} Z_i Z_j}$ implemented through the circuit



with rotation gate $R_Z(-2\theta_{ij}\gamma) = \begin{pmatrix} e^{i\theta_{ij}} & 0 \\ 0 & e^{-i\theta_{ij}} \end{pmatrix}$

variational quantum algorithms

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$$U(\theta) = U_L(\theta_L) \cdots U_2(\theta_2)U_1(\theta_1), \quad U_l(\theta_l) = \prod_m e^{i\theta_m^l H_m^l W_m^l}$$

- its design mostly based on intuition and heuristics
- its structure mostly problem-specific as tailored to the problem structure
- can be problem-agnostic too; e.g., the alternating cost-mixer layering structure for QAOA
- hardware-informed design (at least in near term)
 - short decoherence time, 'localized' operations, small systems
 - need to reduce the depth (and width) of the circuit
 - take unitaries from quantum gates determined by the connectivity/interactions specific to a quantum hardware

- complexity versus expressibility/performance tradeoff
 - parameterized circuit limits the ‘search’ space, more parameters higher capacity
 - constrained by hardware and by trainability (classical computation)
 - ansatz expressibility: if the circuit can be used to uniformly explore the entire space of quantum states

variational quantum algorithms

sampling and expectation

- cost function is often the expectation of a cost Hamiltonian H

$$C(\theta) = \langle \psi_0 | U^\dagger(\theta) H U(\theta) | \psi_0 \rangle$$

- Hamiltonian often well-structured; e.f., k -local
- commutative versus non-commutative terms
 - commutative terms can be measured simultaneously
- $O(\frac{1}{\epsilon^2})$ sample complexity required for given accuracy ϵ
 - $O(\log \epsilon^{O(1)})$ complexity for classical neural networks

variational quantum algorithms

gradient of cost function

- finite difference method

$$\partial_k C(\theta) = \frac{C(\theta + \epsilon \Delta_k) - C(\theta - \epsilon \Delta_k)}{2\epsilon} + O(\epsilon^2)$$

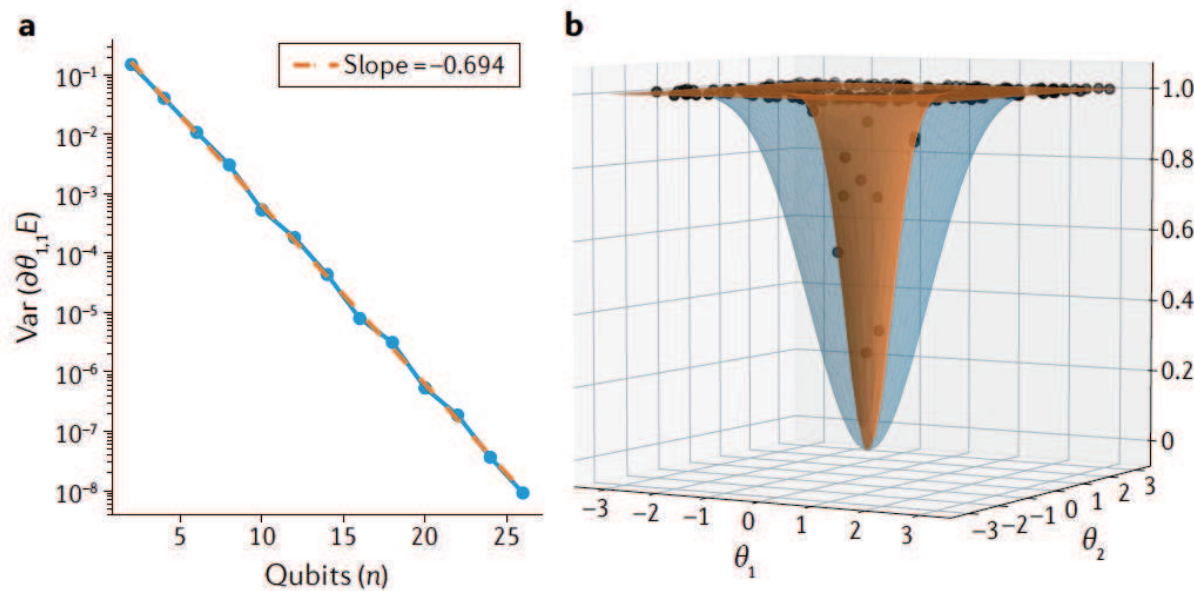
- estimate analytical gradient
 - Hamiltonian with single Pauli terms
 - parameter-shifting method for Hamiltonian with multiple Pauli terms

$$\partial_k C(\eta) = C(\eta + \frac{\pi}{4} \Delta_k) - C(\eta - \frac{\pi}{4} \Delta_k)$$

variational quantum algorithms

trainability

- Barren plateau: regions of cost function landscape where partial gradients vanish exponentially with the size of the system, causing the optimization process to stall



(Cerezo *et al* '21)

variational quantum algorithms

trainability

- barren plateau: regions of cost function landscape where partial gradients vanish exponentially with the size of the system, causing the optimization process to stall
 - analysis on barren plateau: unstructured ansatz, non-local cost functions, etc
 - mitigation strategies: parameter initialization. more structured ansatz

variational quantum algorithms

solution quality

- not sure if it can outperform classical algorithms and under what circumstances it should be used